MIL-T-38049A (USAF)
27 November 1972
SUPERSEDING
MIL-T-38049 (USAF)
14 December 1962

MILITARY SPECIFICATION

TRANSMITTER, PRESSURE TRU-60/A

1. SCOPE

1.1 This specification covers one type of pressure transmitter, designated TRU-60/A.

*2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-T-5350	Transmitters, Synchro, Operated Aircraft, General Specification For
MIL-I-7057	Indicators, Synchro, Aircraft, General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-S-20708	Synchros, General Specification For
MIL-B-27497	Bearing, Jewel, Sapphire or Ruby Synthetic
MIL-I-38121	Indicator, Pressure EGU-9/A

STANDARDS

Military

MIL-STD-143	Specifications and Standards Order of Precedence for the Selection Of
MIL-STD-461	Electromagnetic Interference, Characteristics Requirements For Equipment

FSC: 6620

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MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization Of
MIL-STD-794	Part and Equipment, Procedures For Packaging and Packing Of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-831	Test Reports, Preparation Of
MS33649	Bosses, Fluid Connection - Internal Straight Thread
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 Qualification. Transmitters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.4).

3.2 General specification. The requirements of MIL-T-5350 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials.

3.4.1 Magnesium alloy. Magnesium alloy parts shall not be used.

3.5 Design and construction. The transmitter shall be designed and constructed to operate with an indicator conforming to MIL-I-38121. The transmitter shall be designed to accomplish the following:

a. Sense the engine inlet total pressure (P_{t2})
b. Sense the turbine exhaust total pressure (P_{t7})
c. Divide P_{t7} by P_{t2} and provide a synchro output signal for indication.

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3.5.1 Stops. The mechanism of the transmitter shall include suitable stops that will allow the indicator pointer to register beyond either end of the engine pressure ratio (EPR) range but will protect the mechanism and electronics from overpressure conditions.

*3.5.2 Reliability. Reliability of the transmitter shall be a major design criterion and shall be such as to provide maximum periods of failure free operation. The transmitter shall have a minimum acceptable MTBF of 750 hours as defined per MIL-STD-781, paragraph 3.1. Failure is defined as any malfunction or change in performance of any component that results in degraded performance regardless of the cause (careless handling or involuntary destruction excepted).

3.5.3 Maintainability. The transmitter shall be so designed and constructed that adjustments and repairs can easily be made by the personnel of operating units and overhaul bases without the use of special tools.

*3.5.4 Voltage. The transmitter shall be designed to operate from a 115V 400 Hz single-phase power source in accordance with MIL-STD-704. The transmitter shall meet the functional requirements of MIL-S-20708. The transmitter shall provide the 26V 400 Hz single phase AC input power to the indicator (pin E) in accordance with MIL-STD-704, except the ratio of volts to frequency shall be 26/400Hz $\pm$ 5 percent.

3.5.5 Calibration. The transmitter shall be so calibrated that the rotor of the output synchro transmitter rotates over a range of 330° for a corresponding pressure ratio of 1.0 to 2.5 sensed by the transmitter. The electrical zero setting of the output synchro transmitter incorporated therein shall be such that the transmitter will operate satisfactorily with single synchro style indicators as specified in MIL-I-7057. When increasing pressure ratio values are sensed by the transmitter, the output signals produced shall cause the respective indicator to likewise register compatible increasing readings. The same relationship shall apply for decreasing pressure ratio measurements.

3.5.6 Power consumption. The maximum power consumption shall not exceed 30 VA.

*3.5.7 Bearings Synthetic. When jewel, sapphire or ruby, synthetic bearings are used they shall be in accordance with MIL-B-27497.

3.6 Performance.

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3.6.1 Temperatures. The transmitter shall be capable of operating satisfactorily under the following temperature ranges:

a. -54° to +121°C, operational to specified tolerances.

b. -65° C exposure.

3.6.2 Endurance. The transmitter shall be capable of withstanding 1,500 hours of electrical operation.

*3.6.3 Radio noise suppression. The frequency range of radiated noise shall be 0.10 to 1,000 MHz. When filtering is required, the volume and weight of filtering components shall be minimized by proper electrical and mechanical design. The transmitter shall not be susceptible to electrical interference exceeding the limits defined in MIL-STD-461 and MIL-STD-462.

3.6.4 Dielectric strength. Insulation for electrical circuitry housed in nonhermetically sealed transmitters shall be sufficient to withstand a potential of 1,250V AC rms between the circuitry and the case. Insulation housed within hermetically sealed transmitters shall withstand a potential of 500V AC rms between the circuitry and the case.

3.6.5 Response time. The transmitter response time shall be within the limits specified in Table I. The indicator or recorder shall begin to move within 1/4 second plus the time specified in Table I after the applied pressure is induced.

*3.6.6 Sensitivity. The change in pressure necessary to cause a change of indication shall not exceed the values specified in Table II.

TABLE I

Response Rate

Engine Inlet Pressure (P_{t2}) (Inches Hg Absolute)	Engine Exhaust Pressure (P_{t7}) (Inches Hg Absolute)		Ratio Change	Allowable Time (Seconds)
5.00	6.00	7.00	0.20	1.0
5.00	6.00	12.50	1.30	5.0
15.00	18.00	37.50	1.30	5.0
15.00	25.50	37.50	0.40	2.0
28.00	32.60	70.00	1.34	5.0
42.00	50.40	63.00	0.30	1.5

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Inlet Pressure P_{t2} (Inches Hg Absolute)	Exhaust Pressure P_{t7} (Inches Hg Absolute)	Pressure Ratio	Sensitivity (In. Hg)	
			P_{t2} Constant P_{t7} Varying	P_{t2} Varying P_{t7} Constant
5.00	9.50	1.90	0.008	0.008
11.00	23.10	2.10	0.030	0.030
31.00	65.10	2.10	0.12	0.12

Sensitivity Test Pressure

TABLE II

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3.6.7 Pressure ranges. The engine inlet total pressure (P_{t2}) shall vary from 5 to 45 inches Hg absolute and the exhaust outlet total pressure (P_{t7}) shall vary from 5 to 70 inches Hg absolute. The differential pressure (P_{t7}-P_{t2}) shall range from 0 to 55 inches Hg. A maximum overpressure of 55 inches Hg absolute shall be applied to the P_{t2} and an overpressure of 110 inches Hg absolute to the P_{t7}.

3.7 Case. The case for housing the components of the transmitter shall be fabricated of aluminum. It shall be designed to provide the degree of ruggedness and sealing characteristics required for the transmitter to satisfactorily fulfill the applicable tests specified in section 4. It shall be so designed that the case may be opened, component parts removed and replaced and the case resealed without the use of special tools.

*3.7.1 Case finish. The inside and outside surfaces of the case shall be anodized in accordance with MIL-A-8625. The color shall be natural aluminum. The inside surface of the case may be finished in durable dull black to provide heat-transfer characteristics.

3.8 Damping. Damping shall be incorporated in the transmitter design to prevent excessive oscillation of the output signal. The degree of damping shall be consistent with applicable test requirements specified herein.

3.9 Screw safety-lock. The transmitter external body screws and internal screws shall be secured with safety wire or suitable lock washers. The method selected shall be approved by the procuring activity prior to use.

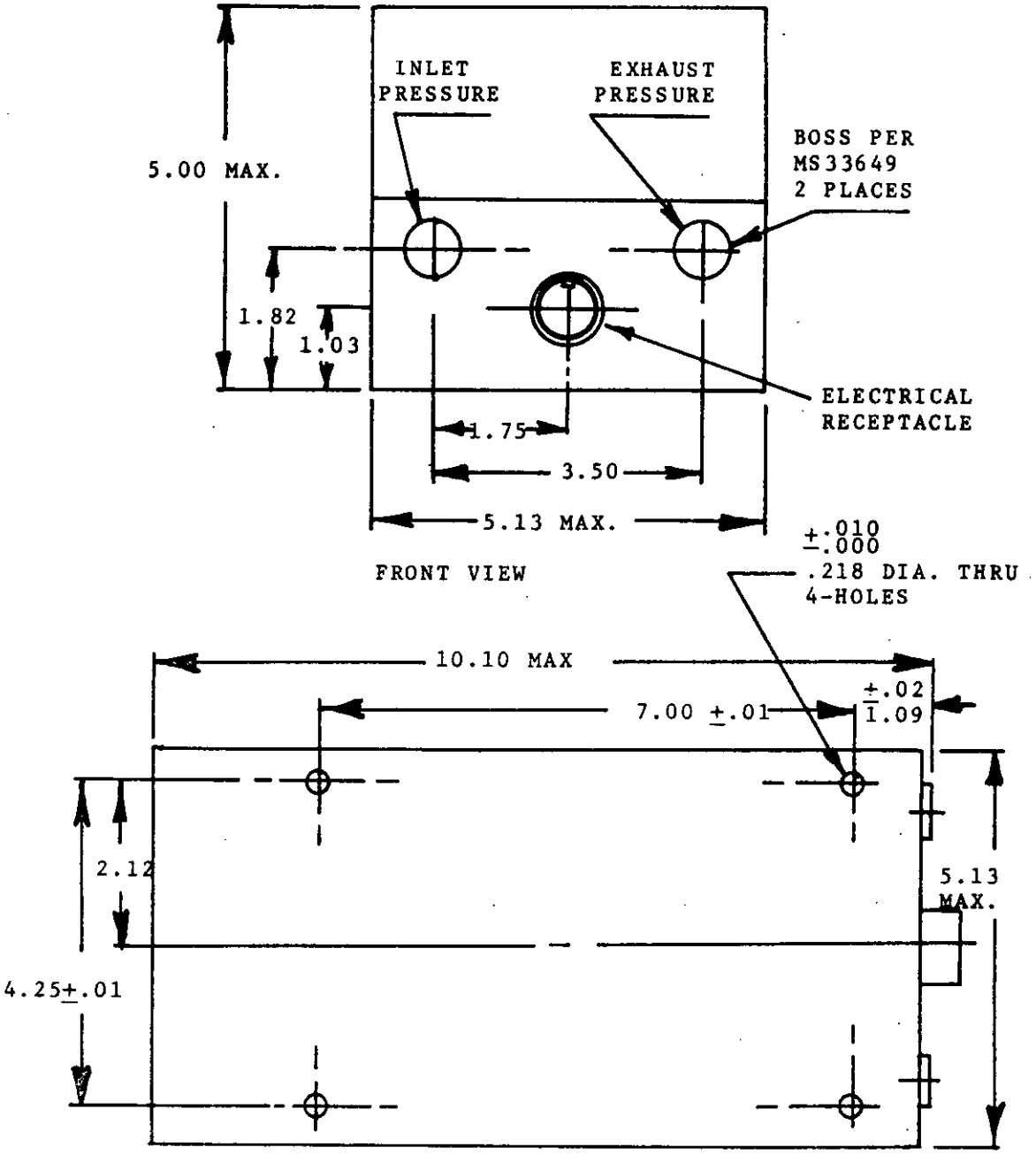
3.10 Mounting. Mounting bosses and the location thereof shall be as shown on Figure 1. A mounting rack incorporating vibration isolators shall be provided. The mounting rack shall be fabricated of aluminum and shall be anodized in accordance with MIL-A-8625. The color shall be natural aluminum.

3.11 Fittings. The P_{t7} and P_{t2} pressure connection fittings and location thereof shall be as shown on Figure 1. Metal tubing shall be used to connect rack fittings to transmitter fittings, if required.

3.12 Caps. The pressure connections shall be provided with caps to protect the threads and prevent the entrance of foreign material. Plugs or tape shall not be used.

3.13 Electrical receptacle. An electrical receptacle as specified on Figure 2 shall be provided.

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BOTTOM VIEW (MOUNTING RACK ONLY)

UNLESS OTHERWISE SPECIFIED,
DIMENSIONS ARE IN INCHES.
TOLERANCES; DECIMALS ±.03

FIGURE 1 OUTLINE DIMENSIONS

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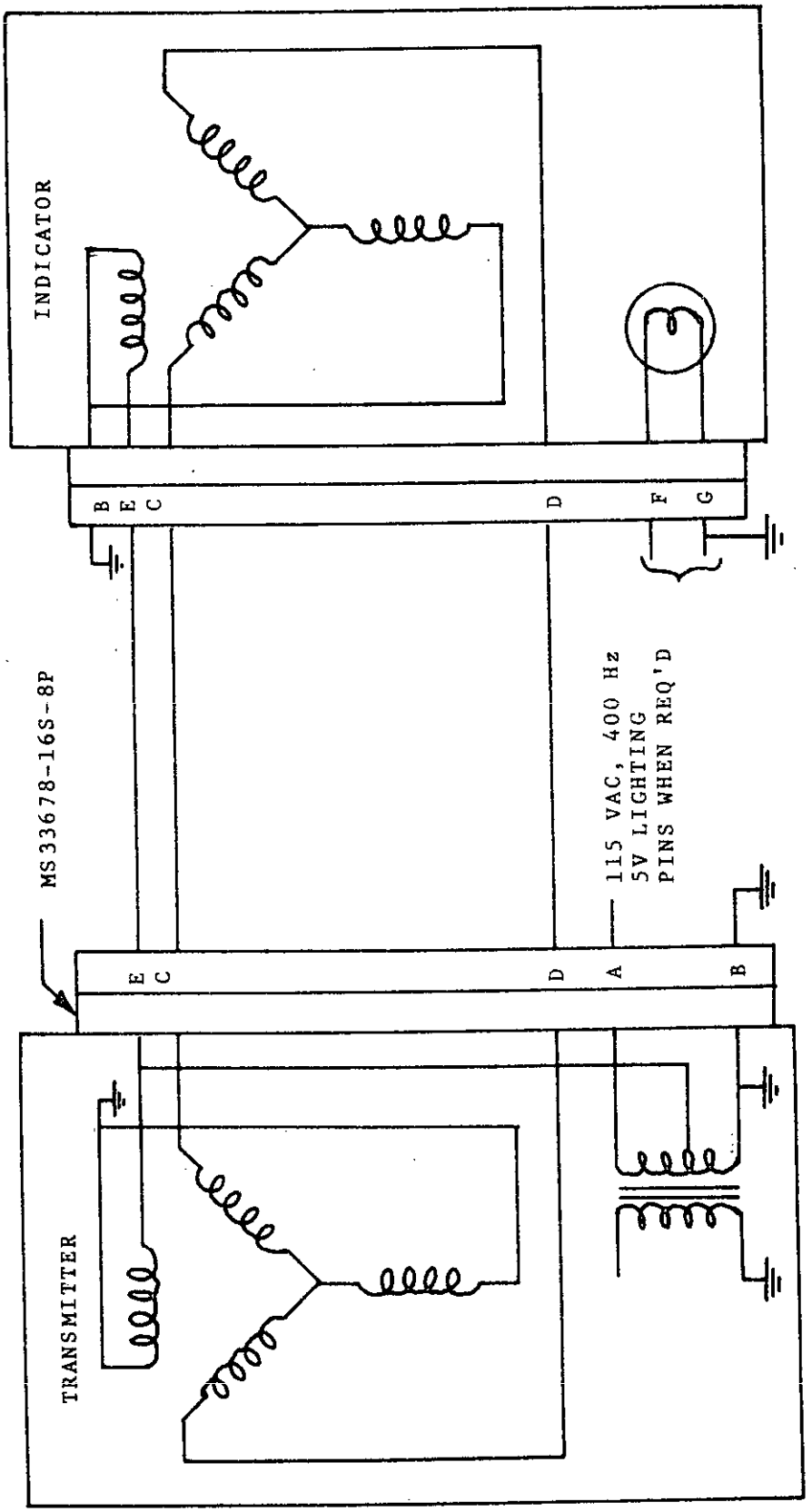


FIGURE 2 WIRING DIAGRAM

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*3.14 Dimensions. The outline dimensions of the transmitter with isolators, including the vibration envelope should not exceed the dimensions specified in Figure 1.

3.15 Weight. The weight of the transmitter shall be kept to a minimum and shall not exceed 5.5 pounds, including shock mount.

3.16 Markings.

3.16.1 Pressure connections. Pressure connection identification shall be permanently and legibly marked as shown on Figure 1.

*3.16.2 Installation diagram. The external wiring diagram (transmitter to indicator) shall be permanently marked on a metal plate which shall be attached to the transmitter by means of screws, rivets, or an approved epoxy adhesive. The wiring diagram shall be in accordance with Figure 2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of transmitter shall be classified as follows:

a. Qualificaiton tests (See para 4.4)
b. Acceptance tests (See para 4.5)

4.3 Test conditions.

4.3.1 Atmospheric. Unless otherwise specified, all tests shall be conducted at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressures or temperatures substantially different from the above values, proper allowance shall be made from the specified condition.

4.3.2 Attitude. Unless otherwise specified, the transmitter shall be tested in normal operating position.

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4.3.3 Test indicator. When a pressure ratio indicator is specified for conducting tests, a synchro indicator standard, as approved by the procuring activity, shall be used.

4.3.4 Test pressure tolerances. Pressure shall be measured to the following accuracies:

Manometer Tolerances (Inches Hg)

0 to 30	30 to 60	30 to 100
±0.0075	±0.0075 increasing to ±0.010	±0.0075 increasing to ±0.015

4.4 Qualification testing.

4.4.1 Test samples. The test samples shall consist of three transmitters representative of the production equipment. Samples shall be identified with the manufacturer's part number and any such other information as required by the procuring activity.

*4.4.2 Test report. After the contractor completes the qualification tests, he shall prepare a test report in accordance with MIL-STD-831 and furnish three complete copies to the procuring activity.

4.4.3 Qualification tests. The qualificaiton tests shall consist of all the tests specified under 4.6.

4.4.3.1 Qualification test schedule. The qualification test schedule shall be as specified in Table III.

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests
b. Sampling plans and tests.

*4.5.1 Individual tests. Each transmitter shall be subjected to the following tests as described under 4.6.

a. Examination of product
b. Leakage.
c. Short scale at room temperature
d. Position error
e. Friction error

4.5.2 Sampling plans and tests.

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Paragraph No.	Test	Test Sample No. 1	Test Sample No. 2	Test Sample No. 3
4.6.1	Examination of product	x	x	x
4.6.2	Leakage	x	x	x
4.6.3	Scale error at room temperature	x	x	x
4.6.4	Position error	x	x	x
4.6.5	Friction error	x	x	x
4.6.6	Response rate	x	x	x
4.6.7	Sensitivity	x	x	x
4.6.8	Overpressure	x	x	x
4.6.9	Dielectric strength			x
4.6.10	Radio noise suppression	x		
4.6.11	Low temperature	x	x	x
4.6.12	High temperature	x	x	x
4.6.13	Extreme low temperature exposure	x		
4.6.14	Extreme high temperature exposure	x		
4.6.15	Vibration			x
4.6.16	Humidity	x		
4.6.17	Fungus			x
4.6.18	Salt spray	x		
4.6.19	Sand and dust			x
4.6.20	Sealing	x		
4.6.21	Endurance and reliability	x		

Transmitter Test Schedule

TABLE III

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4.5.2.1 Sampling Plan A. Three transmitters shall be selected at random from each 100 or fraction thereof produced and subjected to the following tests as described under 4.6:

a. Individual tests
b. Dielectric strength
c. Radio noise suppression
d. Low temperature
e. Scale error at room temperature after low temperature
f. High temperature
g. Scale error at room temperature after high temperature
h. Response rate
i. Sensitivity
j. Overpressure

4.5.2.2 Sampling plan B. Unless otherwise specified, 3 transmitters shall be selected at random from the first 10 items produced and subjected to the following tests:

a. Sampling plan A tests
b. Extreme low temperature exposure
c. Extreme high temperature exposure
d. Vibration
e. Humidity
f. Fungus
g. Salt spray
h. Sand and dust
i. Sealing
j. Endurance and reliability

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no item still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational reasons individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

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4.6 Test methods.

4.6.1 Examination of product. The transmitter shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 Leakage. A pressure of 42 inches Hg absolute shall be applied to the P_{t2} port. The pressure line shall then be sealed upstream of the manometer and with the P_{t7} port open to ambient pressure, the P_{t2} pressure shall not decrease more than 0.05 inch Hg in 5 minutes. A pressure of 70 inches Hg absolute shall be applied to the P_{t7} port. The pressure line shall then be sealed upstream of the manometer and with the P_{t2} port held at approximately 35 inches Hg absolute, the P_{t7} pressure shall not decrease more than 0.05 inch in 5 minutes. The internal volume of the isolated part of the system (manometer wells, pressure lines, et cetera) shall not exceed 45 cubic inches.

*4.6.3 Scale error at room temperature. The transmitter shall be connected to an indicator having a known accuracy and subjected to a scale error at room temperature test at the test points specified in Table IV. The tolerance shall not exceed those specified. The temperature shall be maintained at 25 ± 2°C. During this test, the transmitter shall be vibrated. This test shall be conducted both up scale and down scale.

4.6.3.1 Short scale error. When a short scale error test is specified, the test points shall be those indicated as E in Table IV. Readings before tapping need not be taken unless a friction error test is specified in conjunction with this test.

*4.6.4 Position error. A position error test shall be conducted at 0°, 45°, 90°, 135°, 180°, 225°, and 315° around the major and minor axes. The transmitter shall be subjected to this test only at room temperature and at the points indicated as P in Table IV. The allowable tolerances for the 90°, 135°, 180°, and 225°, positions shall be increased to ± 0.015 EPR for the 1.2 through 2.5 EPR range.

4.6.5 Friction error. A friction error test shall be conducted and the tolerances shall not exceed those specified in Table IV.

4.6.6 Response rate. Pressure shall be applied to the engine inlet pressure connection and the pressure stabilized at each point specified in Table I. Pressure shall then be applied to the engine exhaust pressure connection for the smaller value listed in Table I which corresponds to the engine inlet pressure used. The pressure applied to the engine exhaust pressure connection shall then be raised to and stabilized at the higher pressure specified in Table I which corresponds to the engine inlet pressure used. The time required for the indicator to

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Engine Inlet Pressure (in. Hg Absolute	Engine Exhaust Pressure Inches Hg Absolute								
5.00	5.30	P 6.00	7.00	7.50	E 8.50	S 9.50	10.50	11.50	12.50
7.50	5.95	9.00	E 10.50	11.25	12.75	14.25	15.75	17.25	PE 18.75
11.00	11.66	13.20	15.40	16.50	E 18.70	20.90	S 23.10	E 25.30	27.50
15.00	15.90	18.00	21.00	E 22.50	25.50	28.50	E 31.50	34.50	37.50
20.00	21.20	E 24.00	28.00	30.00	34.00	38.00	42.00	46.00	50.00
25.00	E 26.50	30.00	35.00	EP 37.50	42.50	E 47.50	52.50	E 57.50	62.50
28.00	29.68	32.60	E 39.20	42.00	E 47.60	53.20	E 58.80	64.40	E 70.00
31.00	32.86	E 37.20	43.40	E 46.50	52.70	58.90	S 65.10	71.30	
35.00	37.10	42.00	E 49.00	52.50	59.50	E 66.50			
42.00	44.52	50.40	58.80	63.00	E 71.40				
PRESSURE RATIO	1.06	1.20	1.40	1.50	1.70	1.90	2.10	2.30	2.50

TABLE IV

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P = Position error test points

E = Short scale error test points

S = Sensitivity test points

Scale Error Tolerances for Room Temperature:

1.06 to 1.15 EPR = ± 0.025 EPR
1.2 to 1.4 EPR = ± 0.020 EPR
1.5 thru 2.5 EPR = ± 0.015 EPR

Scale Error Tolerances for Low and High Temperatures, Vibration, and Salt Spray:
1.2 to 1.4 EPR = ± 0.025 EPR
1.5 thru 2.5 EPR = ± 0.020 EPR
1.06 to 1.15 EPR = ± 0.030 EPR

Friction or Position Error Tolerances for Room, Low, and High Temperatures:
1.2 thru 2.5 EPR = ± 0.005 EPR
1.0 to 1.15 EPR = ± 0.007 EPR

4.6.6 cont'd:
stabilize at the reading shall be recorded. This procedure shall be repeated at the same test points but at decreasing engine exhaust pressures.

*4.6.6.1 Response time. The response time during each step shall not exceed the values specified in Table I. The pointer shall be considered stabilized when the overshoot does not exceed 0.010 inch. The pointer shall begin to move within 0.25 second after applying the step input of pressure to the transmitter and shall stabilize within 1-1/2 seconds after it first passes the intended reading.

4.6.7 Sensitivity. The engine inlet and engine exhaust pressures shall be adjusted and stabilized to the pressures specified in Table II. The engine inlet pressure shall be held constant and the engine exhaust pressure increased until a change of indication is detected. The exhaust pressure shall then be decreased slowly until a change of indication is detected. The decrease in pressure necessary to cause the change in indication shall be no greater than the values specified in Table II. This procedure shall be repeated, varying the engine inlet pressure and holding the engine exhaust pressure constant.

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4.6.8 Overpressure. The transmitter shall be subjected to a P_{t2} pressure of 55 ± 1 inch Hg absolute and a P_{t7} pressure of 110 ± 1 inch Hg absolute. The resultant indicated pressure ratio shall be accurate to within ± 0.100 pressure ratio units. The short scale error at room temperature test shall then be conducted and the errors shall not exceed those specified in Table IV.

4.6.9 Dielectric strength. The transmitter shall be tested for dielectric strength. A voltage of 1,250V AC rms at commercial frequency shall be applied to nonhermetically sealed transmitters for 2 seconds and a voltage of 500V AC rms at commercial frequency shall be applied to hermetically sealed transmitters for a minimum period of 5 seconds. Transistors and similar sensitive parts may be disconnected from the circuit prior to testing. There shall be no breakdown of insulation. If the transmitter case is filled with an inert gas and hermetically sealed, the potential shall be reduced to 250V AC rms and the same test conducted.

*4.6.10 Radio noise suppression. The transmitter shall be subjected to transmitted and susceptibility radio noise suppression tests in accordance with MIL-STD-461 and MIL-STD-462.

4.6.11 Low temperature. The transmitter shall be placed in a test chamber and subjected to a temperature of -54° ±2°C for a period of 4 hours. While at this temperature, the short scale error test shall be conducted (see 4.6.3.1). The scale error tolerances shall not exceed those specified in Table IV. At the completion of this test, the transmitter shall be subjected to short scale error and friction error tests at room temperature. The tolerances shall not exceed those specified.

4.6.12 High temperature. The transmitter shall be placed in a test chamber and subjected to a temperature of 100° ±2°C for a period of 4 hours. While at this temperature, the short scale error test shall be conducted (see 4.6.3.1). The scale error tolerances shall not exceed those specified in Table IV. The transmitter shall then be subjected to the short scale error and friction error tests at room temperature. The tolerances shall not exceed those specified.

4.6.13 Extreme low temperature exposure. The transmitter shall be placed in a test chamber and subjected to a temperature of -65° ± 2°C for a period of 48 hours. The temperature shall then be raised to -54° ±2°C and maintained for 24 hours. While still at -54° ±2°C, the transmitter shall be subjected to the short scale error test at all test points specified in Table IV. Scale error tolerances shall not exceed those specified. The transmitter shall be returned to room temperature for a minimum period of 4 hours and subjected to the short scale error test at room temperature test.

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4.6.14 Extreme high temperature exposure. The transmitter shall be placed in a test chamber and subjected to a temperature of 121° ±2°C for a period of 24 hours. Power shall be applied continuously to the transmitter throughout the last 8 hours of this period. While still at 121° ±2°C, the transmitter shall be subjected to a short scale error test. High temperature scale errors shall not exceed the tolerances specified in Table IV. The transmitter shall be returned to room temperature for a minimum period of 4 hours and subjected to the short scale error test at room temperature.

4.6.15 Vibration. The transmitter shall be tested within the range of frequencies and vibratory accelerations specified in MIL-E-5272, procedure XII, Curve A.

4.6.15.1 Resonant frequencies. Function of the transmitter shall be checked concurrently with the operation of scanning the frequency range for resonant frequency. The transmitter shall be vibrated for 1 hour through each of three mutually perpendicular axes of vibration at the resonant condition. When more than one resonant frequency is encountered with vibration applied along any one axis, the test shall be accomplished at the most severe resonance or the test period may be divided among the resonant frequencies, whichever is most likely to produce failure. This test shall be conducted at room temperature. During each resonance endurance period, the transmitter shall be connected to an indicator standard and subjected to a scale error test at the following pressure values. The tolerance for this test shall be 0.025 pressure ratio units.

a. Inlet pressures, inches Hg absolute: 8, 20, and 32
b. Pressure ratios at each inlet pressure: 1.5 and 2.2.

4.6.15.1.1 When resonant frequencies are not apparent within the specified frequency range, the transmitter shall be vibrated for twice the specified time period at a frequency of 55 Hz and an applied double amplitude of 0.060 inch.

4.6.15.2. Scale error after vibration. Following the vibration period, the transmitter shall be tested in accordance with 4.6 and 4.6.3. Allowable tolerances specified in Table IV shall apply.

4.6.16 Humidity. The transmitter, except when completely hermetically sealed, shall be subjected to the humidity test in accordance with procedure I of MIL-E-5272.

4.6.17 Fungus. The transmitter, except when completely hermetically sealed, shall be subjected to the fungus test in accordance with procedure I of MIL-E-5272 for a period of 14 days.

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4.6.18 Salt spray. The transmitter shall be subjected to the salt spray test in accordance with procedure I of MIL-E-5272. Upon completion of the test, the transmitter shall be subjected to the short scale error test specified in 4.6.3.1 at room temperature. The scale errors shall not exceed the room temperature tolerances specified in Table IV.

4.6.19 Sand and dust. The transmitter shall be subjected to a sand and dust test in accordance with procedure I of MIL-E-5272, and then checked for accuracy at room temperature as specified in Table IV at the short scale error test points.

4.6.20 Sealing. The transmitter, if hermetically sealed, shall be tested by means of a mass spectrometer type of helium leak detector. Where a nitrogen-helium mixture is utilized as the filling medium, the detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of approximately 1 atmosphere. Where 100 percent helium is utilized as a filling medium, the leak rate shall not exceed 1.0 micron cubic foot per hour.

*4.6.21 Endurance and reliability. A minimum of three transmitters shall be tested for reliability and each shall be operated for 1500 hours. Seven hundred and fifty hours shall be accomplished while P_{t2} is varied from 6 to 11.5 +0.5, -0.0 inch Hg absolute and return to 6 inches Hg absolute at a rate of 5 $\pm$2 cpm. Upon completion of each 100 hours of operation, P_{t2} shall be varied once from ambient to 11+0.0, -0.5 inch Hg absolute and return to ambient within 10 seconds. During the other 750 hours, P_{t7} shall be varied from ambient to 70 +0.0, -0.5 inch Hg absolute and return to ambient at a rate of 5 $\pm$2 cpm. Upon completion of each 100 hours of operation, P_{t7} shall be varied once from ambient to 70 +0.5, -0.0 inch Hg absolute and return to ambient within 10 seconds. The respective pressure sensing elements not being cycled shall be vented to ambient. The transmitter shall be subjected to a scale error test upon completion of each 250 hours of P_{t2} and P_{t7} cycling operation. The tolerances specified in Table IV shall apply. If a failure occurs during the reliability testing, make the necessary reports, record time before failure and cause and or parts causing failure. Continue testing until 1500 hours on each item is reached. After completion of reliability testing each transmitter shall be required to pass the other qualification tests listed under paragraph 4.4. The MTBF of the transmitter, which shall be at least 7500 hours, shall be determined from the following equation. The contractor shall submit a report to the procuring activity which shall contain test procedures, details of any malfunctions, and the corrective action taken. During cycling, differential pressures shall be kept within specified tolerances.

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$$MTBF = \frac{1,500N}{m}$$

Where N = Number of units tested

m = Number of malfunctions.

*4.7 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

*5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. Reference definition within MIL-STD-794. Transmitters shall be packaged in quantity unit packs of one each in accordance with Method 1A-14 of MIL-P-116. Cushioning between interior and exterior container of a type, density and thickness to insure shock transmission does not exceed peak values, in Gs, established for the transmitter when completed packs are submitted to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Same as Level A above except Method III of MIL-P-116 applies in lieu of Method 1A-14.

5.2 Packing.

5.2.1 Level A. Transmitters packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, weather-resistant class unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping containers shall be of uniform shape and size and of minimum tare and cube consistent with the protection required.

5.2.2 Level B. Transmitters packaged as specified in 5.1.1 shall be packed in domestic class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 shall apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during direct domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

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6. NOTES

6.1 Intended use. The TRU-60/A transmitter covered by this specification is intended for use in transmitting synchro signals proportional to engine pressure ratio measurements on turbofan-type engines.

*6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. If sampling plan B tests are not to be conducted
c. Selection of applicable levels of packaging and packing (See Section 5).

6.3 Definition.

6.3.1 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials or both. A hermetic seal is not intended to include seals accomplished by gaskets or O-rings.

*6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is OCAMA (MMEO) Tinker Air Force Base, Oklahoma 73145 and information pertaining to qualification of products may be obtained from that activity.

NOTE: The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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